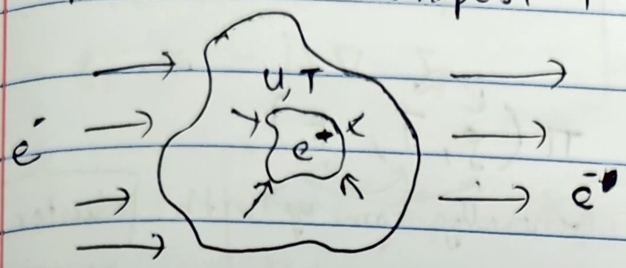
Battery

Mechanics + Transport + Thermal



04/21/22

LECTURE 21

Multiphysics

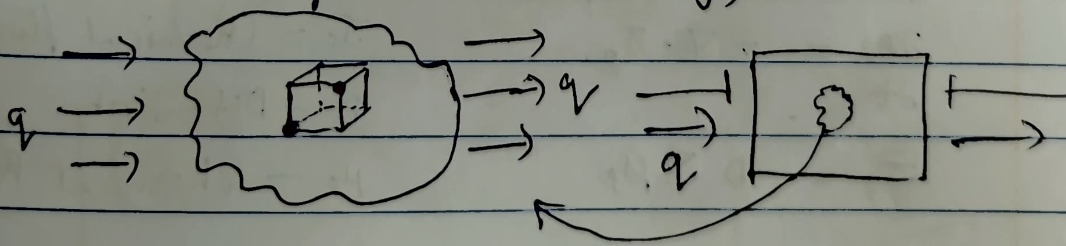
Issues: Existence + uniqueness

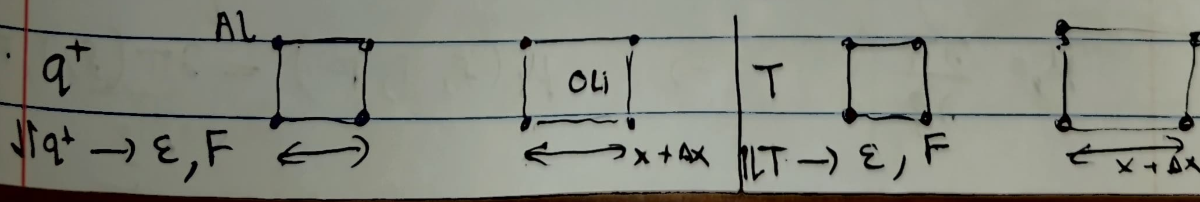
→ Unique solutions
→ Multiple solutions
→ Chaos

Multiphysics Systems

→ Obtained through Variational Arguments
→ Obtained through Phenomenology/ Observation

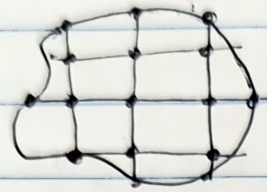
Variational arguments (Battery)





$\uparrow\downarrow q \longrightarrow F \uparrow\downarrow$?

$\uparrow\downarrow T \longrightarrow F \uparrow\downarrow$?



$\Pi(\rho, T, F)$

Generally come up with $\boxed{\text{cluster expansion}}$ in DFT

$\Pi(\rho, T, F)$

Mechanics

1st P-K $\quad P = \dfrac{\partial \Pi}{\partial F}$

(equilibrium) $\quad \nabla \cdot P(F, \rho, T) = 0 \quad -$ (1)

Thermal Trasport

$$\frac{\partial T}{\partial t} = -\nabla \cdot J_T$$

$$\bar{J}_T = -D \nabla \mu_T$$

$$\mu_T = \frac{\partial \Pi}{\partial T}$$

where μ - Thermal potential

D - conductivity

(limit of Reversible Thermodynamics)

$$\mu_T(T, \rho, F) = \frac{\partial \Pi(T, \rho, F)}{\partial T}$$

Species Transport

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot J_\rho$$

$$\bar{J}_\rho = -D \nabla \mu_\rho$$

J_ρ - Chemical flux

D - Diffusivity

μ_ρ - chemical Potential

$$\mu_\rho = \frac{\partial \Pi}{\partial \rho} \quad \rightarrow \quad \mu_\rho(\rho, T, F) = \frac{\partial \Pi(\rho, T, F)}{\partial \rho}$$

$$\left.\begin{aligned} \nabla \cdot P(\rho, T, F) &= 0 \\ T_{,t} &= -D\, \nabla \cdot J_T \\ \rho_{,t} &= -D\, \nabla \cdot J_\rho \end{aligned}\right\} (S) \rightarrow (W) \longrightarrow (W^h) \rightarrow (M)$$

$$\begin{bmatrix} K_{\rho\rho} & K_{\rho T} & K_{\rho u} \\ K_{T\rho} & K_{TT} & K_{Tu} \\ K_{u\rho} & K_{uT} & K_{uu} \end{bmatrix} \begin{bmatrix} \rho \\ T \\ u \end{bmatrix} = \begin{bmatrix} -R_\rho \\ -R_T \\ -R_u \end{bmatrix}$$

Phenomenology / Observations

$$\frac{\partial x}{\partial t} = a + bx + cy + dz + e\,yz$$

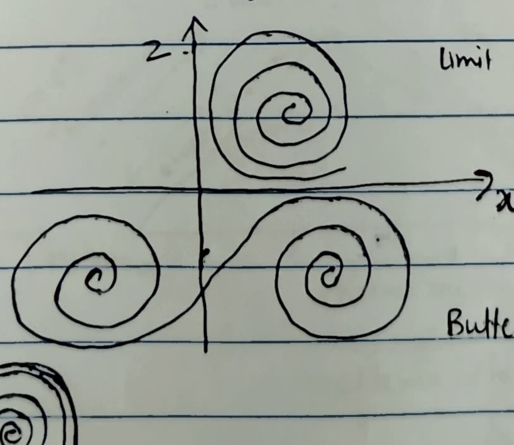
Lorenz (Attractors)

$$\frac{\partial y}{\partial t} = f + gx + hy + iz + jxz$$

Volterra

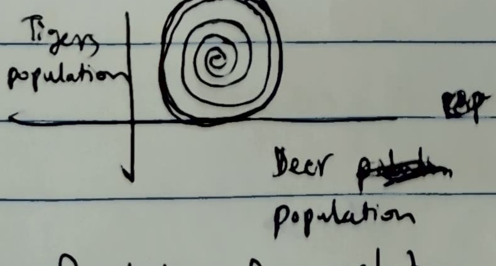
$$\frac{\partial z}{\partial t} = k + lx + my + nz + o\,xy$$

Examples

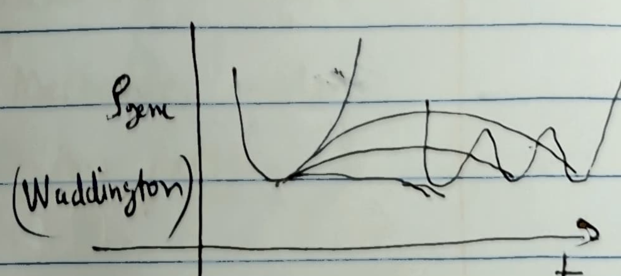


Limit cycle

Butterfly effect



Predator Prey plot



$$\frac{\partial x}{\partial t} = ax + by$$

$$\frac{\partial y}{\partial t} = cx + dy$$

$$(S) \begin{cases} \dfrac{\partial \rho^1}{\partial t} = f(\rho^1, \rho^2) + D \nabla^2 \rho^1 + \mu \, u \cdot \nabla \rho \\ \dfrac{\partial \rho^2}{\partial t} = g(\rho^1, \rho^2) + D \nabla^2 \rho^2 + \mu^2 \, u \cdot \nabla \rho^2 \\ \dfrac{Du}{Dt} = \dfrac{\partial u}{\partial t} + u \cdot \nabla u = \underbrace{-\nabla p(\rho^1, \rho^2) + \nabla \tau}_{\nabla \cdot \sigma} + g \end{cases}$$

Navier's Stokes

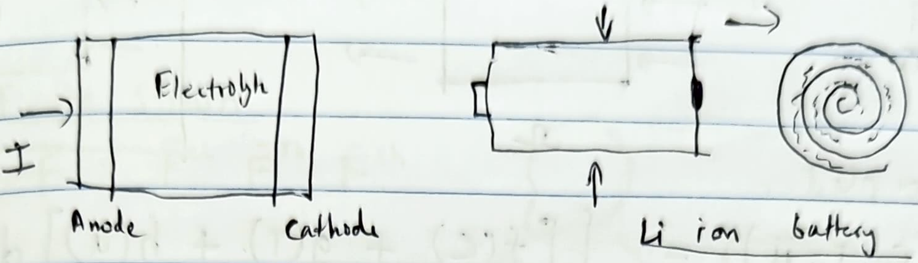
$$(S) \longrightarrow (W) \longrightarrow (W^h) \longrightarrow (M)$$

Generally ill conditioned because they are not coming from variational argument

04/25/22

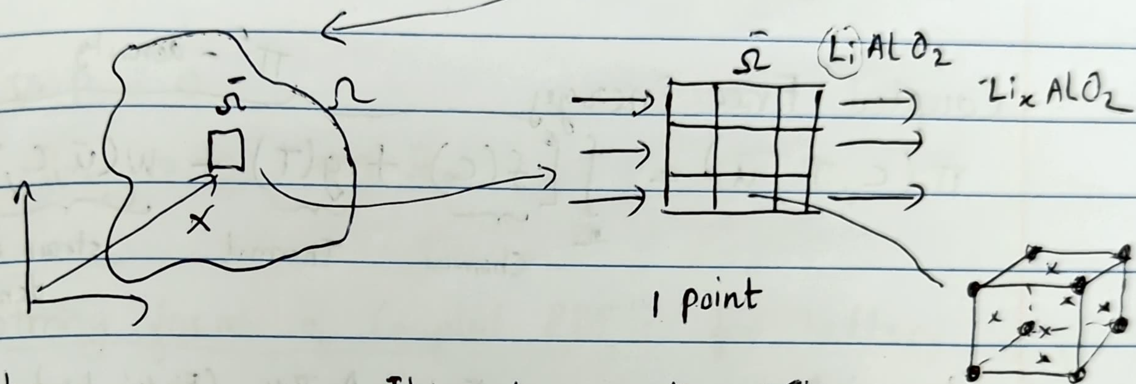
LECTURE 22

Multiphysics of Lithium-ion Batteries



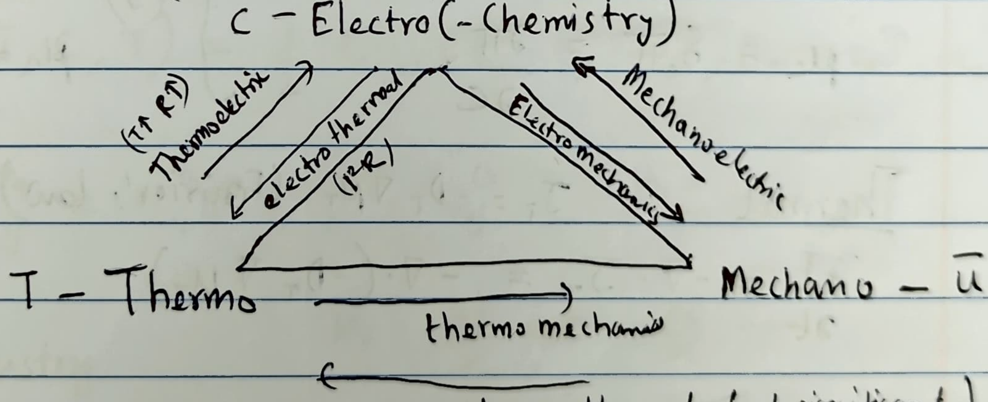
~~Old st~~
Regular batteries

cathode and electrolyte are diffuse
anode



- Charge (Electrochemistry)
- Thermal
- Mechanics

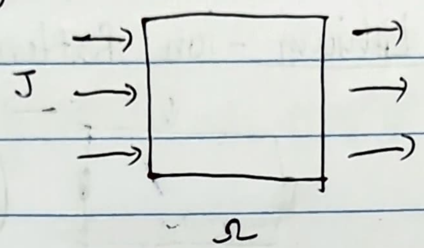
$$\text{Power} = VI = I^2 R = \left(\frac{\partial C}{\partial t}\right)^2 R$$

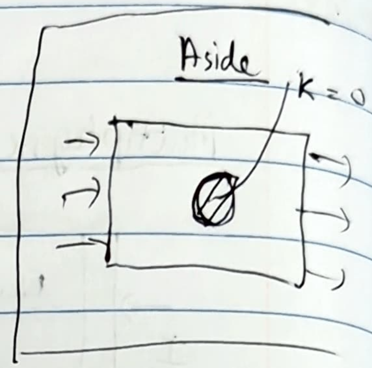


Electro-Thermo-Mechanics model of the Li-ion Battery

Geometry

(fields) c, T, $\bar{u}$





Decoupled

$$\Pi(c, T, \bar{u}) = \int_\Omega \left[f(c) + g(T) + h(\bar{u}) \right] dV$$

Fully decoupled representation

Coupled Free Energy

$$\Pi(c, T, \bar{u}) = \int_\Omega \overbrace{\left[\underbrace{f(c)}_{\text{Chemical}} + \underbrace{g(T)}_{\text{Thermal}} + \underbrace{w(\bar{u}, c, T)}_{\text{strain energy density}} \right]}^{\Pi' \text{ - density}} dV$$

Chemistry

$J_c = -D_c \nabla \mu_c$ (Fick's law)

$$\left.\begin{aligned} \frac{\partial c}{\partial t} &= -\nabla \cdot J_c = -\nabla \cdot (-D_c \nabla \mu_c) \\ \mu_c &= \delta_c \Pi = \frac{\partial \Pi}{\partial c} \end{aligned}\right\} (\bar{S})$$

J_c - Chemical Flux

D_c - Chemical Diffusivity

μ_c - Chemical Potential

Thermal

$J_T = -D_T \nabla \mu_T$ (Fourier's law)

$$\frac{\partial T}{\partial t} = -\nabla \cdot J_T = -\nabla \cdot (-D_T \nabla \mu_T)$$

$$\mu_T = \delta_T \Pi = \frac{\partial \Pi}{\partial T}$$

J_T - Thermal Flux

D_T - Thermal conductivity

μ_T - Thermal potential

Mechanics (Mechanical Growth)

Small Strains

$$\underset{\text{sym}(\nabla \bar{u})}{\bar{\varepsilon}} = \bar{\varepsilon}^{el} + \underset{(\alpha_T \Delta T)}{\bar{\varepsilon}^{th}} + \underset{(\alpha_c \Delta c)}{\bar{\varepsilon}^{ch}}$$

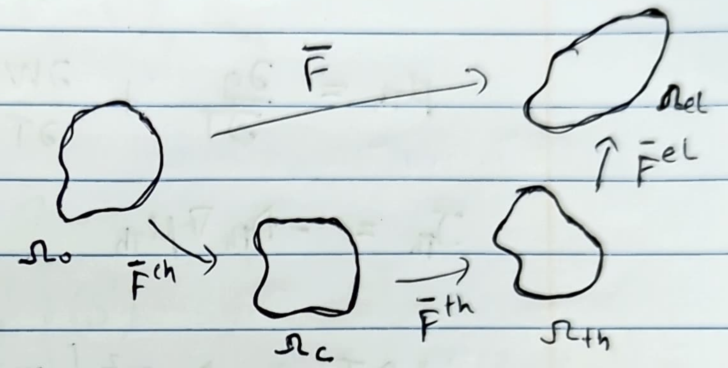
$$\Rightarrow \boxed{\left(\mathbb{C}_{ijkl}\left[\varepsilon_{kl} - \varepsilon_{kl}^{th} - \varepsilon_{kl}^{ch}\right]\right)_{,j} = 0}$$

$$\Rightarrow \left(\mathbb{C}_{ijkl}\,\varepsilon_{kl}^{el}\right)_{,j} = 0$$

$$\Rightarrow \sigma_{ij,j} = 0 \quad \text{on } \Omega$$

<u>Finite strain</u>

$$\bar{F} = \bar{F}^{el}\,\bar{F}^{th}\,\bar{F}^{ch}$$

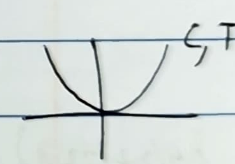


$$\bar{P} = \frac{\partial \Pi}{\partial \bar{F}^{el}}(C, T, \bar{u})$$

$$\therefore \nabla \cdot P = 0$$

$$P_{iJ,J} = 0 \quad \text{on } \Omega_0$$

<u>Strong form of Coupled PDE's for battery</u>



<u>Consider</u>

$$f(c) = \frac{c^2}{2}, \quad g(T) = \frac{T^2}{2}$$

(Small Strain) $$W(\bar{u}, c, T) = \frac{1}{2}\mathbb{C}_{ijkl}\left(\varepsilon_{ij} - \varepsilon_{ij}^{Th} - \varepsilon_{ij}^{ch}\right)\left(\varepsilon_{kl} - \varepsilon_{kl}^{Th} - \varepsilon_{kl}^{ch}\right)$$

where

$$\varepsilon_{ij}^{Th} = \alpha_{ij}^{Th}\Delta T, \quad \varepsilon_{ij}^{ch} = \alpha_{ij}^{ch}\Delta C$$

$$\mu_c = \frac{\partial f}{\partial c} + \frac{\partial W}{\partial c} =$$

<u>Chemistry</u>

$$\frac{\partial c}{\partial t} = D_c \nabla \cdot \nabla \mu_c = D_c \nabla \cdot \nabla\left[C + \frac{\partial W}{\partial c}\right]$$

$$\left.\begin{aligned} &\Rightarrow \frac{\partial c}{\partial t} = D_c \nabla^2\left[C + \frac{\partial W}{\partial c}\right] \quad \text{on } \Omega \\ &C = C_0 \quad \text{on } \partial\Omega_y \\ &\nabla \mu_c \cdot n = J_c \quad \text{on } \partial\Omega_h \end{aligned}\right\} (S^{ch})$$

Thermal

$$\mu_{Th} = \frac{\partial \Pi'}{\partial T} = \frac{\partial}{\partial T}(f + g + w)$$

$$\mu_{Th} = \frac{\partial g}{\partial T} + \frac{\partial W}{\partial T} = T + \frac{\partial W}{\partial T}$$

$$J_{Th} = -D_{Th} \nabla \mu_{Th}$$

$$\left.\begin{aligned} \therefore \frac{\partial T}{\partial t} &= D_T \nabla^2 \left[T + \frac{\partial W}{\partial T} \right] \quad \text{on } \Omega \\ T &= T_0 \quad \text{on } \partial\Omega_g \\ \nabla \mu_T \cdot \bar{n} &= J_{Th} \quad \text{on } \partial\Omega_h \end{aligned}\right\} (S^{Th})$$

04/27/22 LECTURE 23

Coupled PDE's for Li-ion Battery

$$\Pi'(c, T, \bar{u}) = \frac{1}{2} c^2 + \frac{1}{2} T^2 + \frac{1}{2} \mathbb{C}_{ijkl} \varepsilon^{el}_{ij} \varepsilon^{el}_{kl} -$$

$$\bar{\varepsilon}^{el} = \bar{\varepsilon} - \bar{\varepsilon}^{Th} - \bar{\varepsilon}^{ch}$$

$$\bar{\varepsilon}^{T} = \bar{\alpha}_T \Delta T = \bar{\alpha}_T (T - T_{ref}) = \bar{\alpha}_T T \qquad T_{ref} = 0$$

$$\bar{\varepsilon}^{c} = \bar{\alpha}_c \Delta c = \bar{\alpha}_c (c - c_{ref}) = \bar{\alpha}_c c \qquad c_{ref} = 0$$

Strong Form of PDE's

Composition

$$\left.\begin{aligned} c_{,t} &= D^c \nabla \cdot \nabla \mu^c \\ c &= c_0 \quad \text{on } \partial\Omega_g \\ \nabla \mu^c \cdot n &= h^c_0 \quad \text{on } \partial\Omega_h \end{aligned}\right\} (S^c) \qquad \mu^c = \frac{\partial \Pi}{\partial c}$$

$$\mu^c = \mathbb{C} + \mathbb{C}_{ijkl}(\bar{\varepsilon} - \bar{\varepsilon}^c - \bar{\varepsilon}^T)_{ij}\underbrace{\left(-\frac{\partial \varepsilon^c}{\partial c}\right)_{kl}}_{-\bar{\alpha}_c}$$

$$\mu^c = c - \mathbb{C}_{ijkl}(\bar{\varepsilon} - \bar{\varepsilon}^c - \bar{\varepsilon}^T)_{ij}\,\bar{\alpha}_{\varepsilon_{kl}}$$

Temperature

$$\left.\begin{aligned} T_{,t} &= D^T \nabla\cdot\nabla\mu^T \\ T &= T_0 \quad \text{on } \partial\Omega_g^T \\ \nabla\mu^T\cdot n &= h_0^T \quad \text{on } \partial\Omega_h^T \end{aligned}\right\}(S^T)$$

$$\mu^T = \frac{\partial \pi}{\partial T} = T - \mathbb{C}_{ijkl}(\bar{\varepsilon} - \bar{\varepsilon}^c - \bar{\varepsilon}^T)_{ij}\,\bar{\alpha}_{T_{kl}}$$

Mechanics

$$\nabla\cdot\sigma = 0$$

$$\sigma_{ij,j} = 0, \quad \text{where } \sigma = \frac{\partial \pi}{\partial \varepsilon}$$

$$\Rightarrow \sigma_{ij} = \frac{\partial \pi}{\partial \varepsilon_{ij}} = \mathbb{C}_{ijkl}(\varepsilon - \varepsilon^c - \varepsilon^T)_{kl}$$

$$\left.\begin{aligned} \sigma_{ij,j} &= 0 \\ u &= u_0 \quad \text{on } \partial\Omega_g^u \\ \sigma\cdot n &= h_0^u \quad \text{on } \partial\Omega_h^u \end{aligned}\right\}(S^u)$$

$$(S) \longrightarrow (W) \qquad \begin{aligned} w^c,\ c &\in S^c \subset H^1(\Omega) \\ w^T,\ T &\in S^T \subset H^1(\Omega) \\ w^u,\ u &\in S^u \subset H^1(\Omega) \end{aligned}$$

Composition

$$\int_\Omega w^c c_{,t}\, dV = \int_\Omega w^c D^c \nabla\cdot\nabla\mu^c\, dV$$

$$\int w^c c_{,t}\, dV = \int_\Omega D^c \nabla\cdot(w^c \nabla\mu^c)\, dV$$

$$- \int \nabla w^c \cdot D^c \nabla \mu^c\, dV$$

$$\Rightarrow \int_\Omega w^c c_{,t}\, dV = \int_{\partial\Omega_h} D^c\, w^c \nabla\mu^c \cdot n\, dS - \int_\Omega \nabla w^c D^c \nabla\mu^c dV$$

$$\Rightarrow \int_\Omega w^c c_{,t}\, dV + \int_\Omega \nabla w^c D^c \nabla\mu^c\, dV = \int_{\partial\Omega_h} w^c J^c\, dS$$

$$(w^c)$$

$$\nabla\mu^c = \nabla\left[c - \mathbb{C}_{ijkl}(\varepsilon - \varepsilon^c - \varepsilon^T)_{ij}\, \alpha_{c_{kl}}\right]$$

$$= \nabla c - \mathbb{C}_{ijkl}(\nabla\varepsilon - \nabla\varepsilon^c - \nabla\varepsilon^T)_{ij}\, \alpha_{c_{kl}}$$

$$= \nabla c - \mathbb{C}_{ijkl}(0 - \alpha^c \nabla c - \alpha^T \nabla T)_{ij}\, \alpha_{c_{kl}}$$

$$\Rightarrow R^c = 0 \quad (w^c)$$

<u>Temperature</u>

$$R^T = \int_\Omega w^T T_{,t}\, dV + \int_\Omega \nabla w^T \cdot D^T \nabla\mu^T\, dV - \int_{\partial\Omega_{h^T}} w^T J^T dS$$

$$R^T = 0 \qquad (w^T)$$

where $\nabla\mu^T = \nabla T - \mathbb{C}_{ijkl}(0 - \alpha^c \nabla c - \alpha^T \Delta T)_{ij}\, \alpha_{T_{kl}}$

Mechanics

$$\int_{\Omega} w_i^u \sigma_{ij,j} \, dV = 0$$

$$\Rightarrow \int_{\Omega} w^u_{i,j} \sigma_{ij} \, dV = \int_{\partial\Omega_h^u} w h^u \, dS$$

h^c - surface tractions

$$\Rightarrow R^u = \int_{\Omega} w_{i,j} \sigma_{ij} \, dV - \int_{\partial\Omega_h^u} w h^u \, dS$$

$$R^u = 0 \qquad (W^u) \qquad \sigma_{ij} = \mathbb{C}_{ijkl}(\varepsilon_{kl} - \varepsilon^c_{kl} - \varepsilon^T_{kl})$$

Weak form

$$\hat{W} = W^c \cup W^T \cup W^u$$

where $\hat{u} = \begin{Bmatrix} c \\ T \\ \hat{u} \end{Bmatrix}$ primal field

$$\hat{R} = R^c + R^T + R^u = 0 \quad \text{Residual}$$

Solve

$$\boxed{\hat{R} = 0}$$

Solution

$\rightarrow$ Monolithic Scheme

$$\begin{bmatrix} R^c \\ R^T \\ R^u \end{bmatrix}$$

$$\hat{R} = 0$$

$$\Rightarrow \hat{R}^n + \left.\frac{\partial \hat{R}}{\partial \hat{u}}\right|_{i,n} {}^{i}\Delta\hat{u}^n$$

$$\Rightarrow {}^{i}\Delta\hat{u}^n = -\left(\frac{\partial \hat{R}}{\partial \hat{u}}\right)^{-1} ({}^{i}\hat{R}^n)$$

$$\Rightarrow {}^{i+1}\hat{u}^n = {}^{i}\hat{u}^n + \Delta {}^{i}\hat{u}^n$$

Standard Newton Raphson for multi field

$$\left.\frac{\partial \hat{R}}{\partial \hat{u}}\right|_{i,n} = \begin{bmatrix} \frac{\partial R^c}{\partial c} & \frac{\partial R^c}{\partial T} & \frac{\partial R^c}{\partial u} \\ \frac{\partial R^T}{\partial c} & \frac{\partial R^T}{\partial T} & \frac{\partial R^T}{\partial u} \\ \frac{\partial R^u}{\partial c} & \frac{\partial R^u}{\partial T} & \frac{\partial R^u}{\partial u} \end{bmatrix}$$

Solution problem for Monolithic scheme

$$\hat{R} = 0$$

$$\Delta^{i}\hat{u}^{n} = -{}^{i}\hat{K}^{n^{-1}}\left({}^{i}\hat{R}^{n}\right)$$

$${}^{(i+1)}\hat{u}^{n} = {}^{i}\hat{u}^{n} + \Delta^{i}\hat{u}^{n}$$ until tolerance $= |{}^{i}\hat{R}^{n}| < tol$

You need

$$\hat{R},\ \hat{K} = \begin{bmatrix} K_{CC} & K_{CT} & K_{Cu} \\ K_{TC} & K_{TT} & K_{Tu} \\ K_{uC} & K_{uT} & K_{TT} \end{bmatrix}$$

How to find $\left(\hat{K}^{-1}\right)$

large condition number

- Conditioning / Well-Posedness of the coupled PDE's

$$K\hat{u} = \hat{R}$$

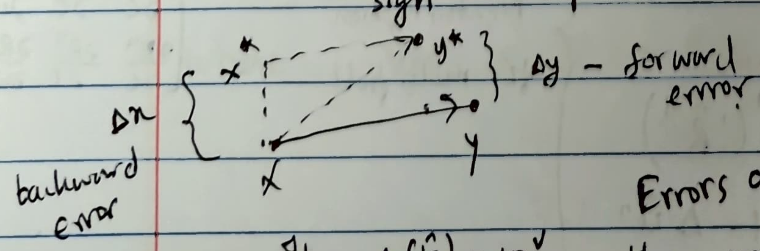
How accurately will this be solved by any solver in a finite precision ~~comp~~ computer

Fixed Precision representation

e.g. double precision → 64 bit

$N \simeq$ __ ____ ________

1 bit	10 bit	53 bit
↓	↓	↓
sign	exponent	mantissa (Decimal places)



Errors $\propto$ cond$(\hat{K}) = \left(\frac{\lambda_{max}}{\lambda_{min}}\right)$ – eigenvalues

If cond$(\hat{K}) = 10^{v}$, then you loose v digits accuracy

05/03/22

LECTURE 24

$$R = 0 \Rightarrow \begin{Bmatrix} R_c \\ R_T \\ R_{\bar{u}} \end{Bmatrix} = 0 \qquad \Delta\hat{u}^I = -\left(\frac{\partial R}{\partial \hat{u}}\right)^{-1} R^I$$

$$\hat{u} = \begin{Bmatrix} c \\ T \\ \bar{u} \end{Bmatrix}$$

$$\frac{\partial R}{\partial \hat{u}} = \begin{bmatrix} R_{cc} & R_{cT} & R_{c\bar{u}} \\ R_{Tc} & R_{TT} & R_{T\bar{u}} \\ R_{\bar{u}c} & R_{\bar{u}T} & R_{\bar{u}\bar{u}} \end{bmatrix}_{3\times 3} \qquad R_{cc} = \frac{\partial R_c}{\partial c} \qquad R_{c\bar{u}} = \frac{\partial R_c}{\partial \bar{u}}$$

Solvability

$$\text{cond}(K) = \frac{\lambda_{max}}{\lambda_{min}}, \qquad \text{cond}(K) \simeq 10^{+v}$$

v – order of precision in the scheme

$v \propto \frac{1}{h^2}$ for poisson problem

Preconditioning

$$\Delta u = K^{-1}(R)$$

$$\Rightarrow K(\Delta u) = R$$

$$\Rightarrow \boxed{Ax = b}$$

If $\text{cond}(A) \gg 1$, then consider using preconditioner

Left Preconditioning

$$Ax = b$$

There exists $P \simeq A^{-1}$ $\qquad \| P - A^{-1} \|_n \leq tol$

Then

$$\underbrace{PA}_{B} x = \underbrace{Pb}_{y}$$

$$Bx = y \Rightarrow x = B^{-1} y$$

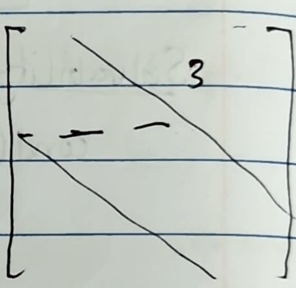
Examples

1) Jacobi Preconditioner

$$P_{Jacobi} = \begin{bmatrix} 1/A_{11} & & & \\ & 1/A_{22} & & \\ & & \ddots & \\ & & & 1/A_{nn} \end{bmatrix}$$

2) ILU – Incomplete LU factorisation

$A = LU$, $A \approx \bar{L}\bar{U}$ (iLU)



3) AMG/GMG (Multigrid method)

Right Preconditioners

$Ax = b$, $\underbrace{AP^{-1}}_{B}\underbrace{Px}_{y} = b$

$By = b \Rightarrow y = B^{-1}b \rightarrow \boxed{x = P^{-1}y}$

STAGGERED SCHEMES

$R_c = 0 \Rightarrow \int_\Omega w\, c_{,t}\, dV + \int_\Omega \nabla w \cdot \nabla c\, dV$

$R_T = 0 \Rightarrow$ similar $+ \int \nabla w(\varepsilon - \varepsilon^T - \varepsilon^c)\, dV = 0$

$R_G = 0 \Rightarrow \int_\Omega \nabla w \cdot \sigma\, dV = 0$

At time T^n (n^{th} increment)

$\rightarrow$ At i^{th} iteration ${}^{i}T^{n}$

$\longrightarrow$ Solve ${}^{i}R_c^n = 0$ for ${}^{i+1}C^n$

$\longrightarrow$ Solve ${}^{i}R_T^n = 0$ for ${}^{i+1}T^n$

but use

$C = {}^{i+1}C^n$

$T = {}^{i}C^n$

$\bar{u} = {}^{i}\bar{u}^n$

$\rightarrow$ Solve ${}^{i}R_{\bar{u}}^n = 0$, but use

$C = {}^{i+1}C^n$

$T = {}^{i+1}T^n$

$\bar{u} = {}^{i}u^n$

Time Integration in Residuals

$$R_c = \int_\Omega w_c C_{,t}\, dV + \int_\Omega \nabla w_c \nabla c\, dV = 0$$

$$R_T = \int_\Omega w_T T_{,t}\, dV + \int_\Omega \nabla w_T \nabla T\, dV = 0$$

$$R_{\bar{u}} = \int_\Omega \nabla w_{\bar{u}} \cdot \sigma\, dV = 0$$

Forward Euler

$$\left.\frac{\partial C}{\partial t}\right|_{i+1} = \frac{{}^{i+1}C^n - {}^{0}C^n}{\Delta t}, \quad \left.\frac{\partial T}{\partial t}\right|_{i+1} = \frac{{}^{i+1}T^n - {}^{0}T^n}{\Delta t}$$

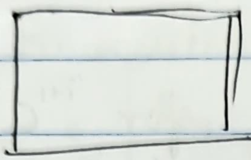
$${}^{i+1}R_c^n = \int_\Omega w_c\left(\frac{{}^{i+1}C^n - {}^{0}C^n}{\Delta t}\right) dV + \int_\Omega \nabla w_c \nabla {}^{0}C^n\, dV = 0$$

$$\Rightarrow \int_\Omega \frac{w_c C}{\Delta t} = \int_\Omega \nabla w_c\, {}^{0}\nabla C^n dV + \int_\Omega \frac{w_c\, {}^{0}C^n}{\Delta t} dV$$

$\rightarrow \quad MC = R \quad \rightarrow {}^{i+1}C^n = M^{-1}R$

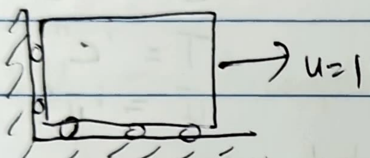
Li; Sn O m.

<u>HW</u>

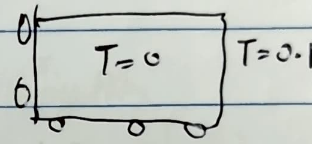


$$\hat{u} = \begin{Bmatrix} c \\ T \\ \bar{u} \end{Bmatrix}$$

1) Only ~~mechanics~~ mechanics

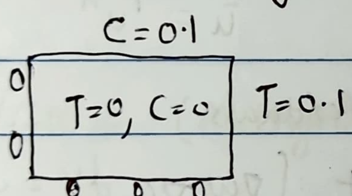


2) Add temperature $(T, \bar{u})$



100 time steps

3) Add Electro- Chemistry $(c, T, \bar{u})$

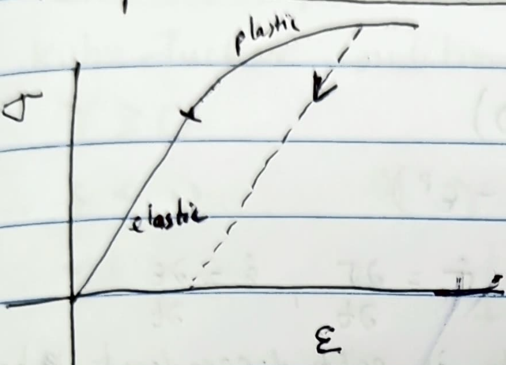


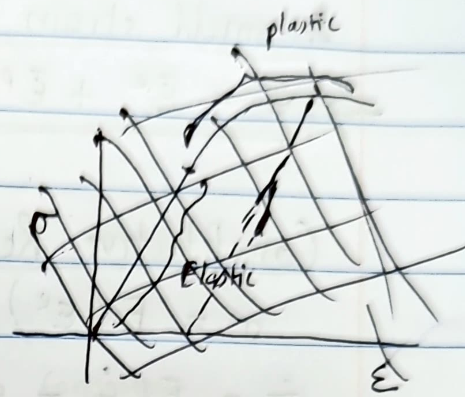
100 time steps

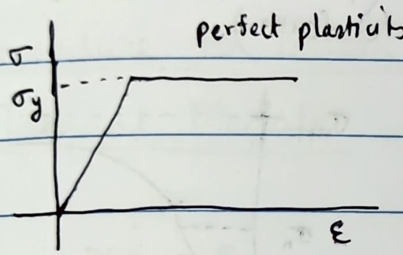
05/05/2022

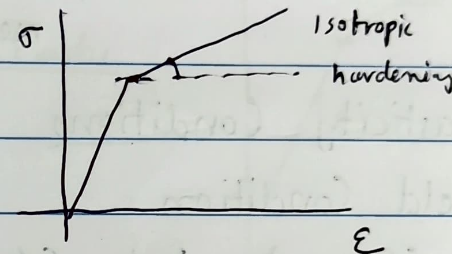
LECTURE 25

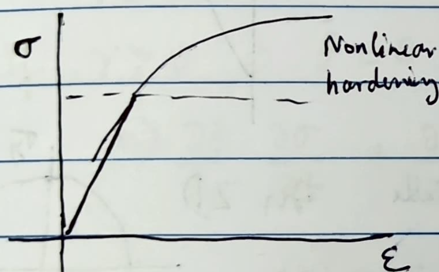
Computational Plasticity

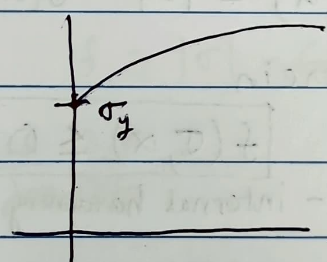








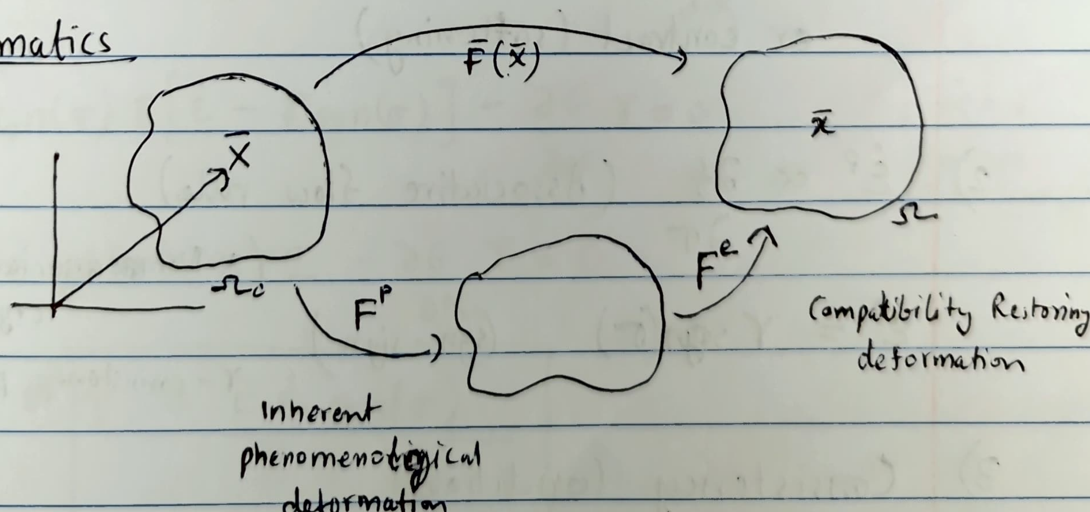




Ramberg
Osgood Relation

$$\sigma = \sigma_y + C\varepsilon_p^{\,n}$$ $n \rightarrow$ Hardening coefficient

Kinematics



$\bar{x} = \bar{F}\bar{X}$ where $\boxed{\bar{F} = \bar{F}^e \bar{F}^p}$ (Multiplicative split of deformation gradient)

Kinematics

$\rightarrow \boxed{\bar{F} = \bar{F}^e \bar{F}^p}$

In small strain limit

$\rightarrow \boxed{\bar{\varepsilon} = \bar{\varepsilon}^e + \bar{\varepsilon}^p}$ (Additive split in small strains)

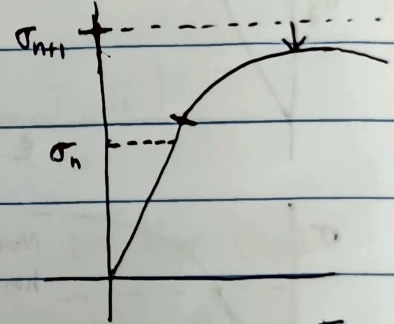
Constitutive Relation (1D)

$$\sigma = E(\varepsilon^e) = E(\varepsilon - \varepsilon^p)$$

$$\dot{\sigma} = E(\dot{\varepsilon} - \dot{\varepsilon}^p) \qquad \dot{\sigma} = \frac{\partial \sigma}{\partial t}, \quad \dot{\varepsilon} = \frac{\partial \varepsilon}{\partial t}$$

(Plasticity is path dependent, hence it needs a rate form)

Plasticity Conditions



1) Yield Condition

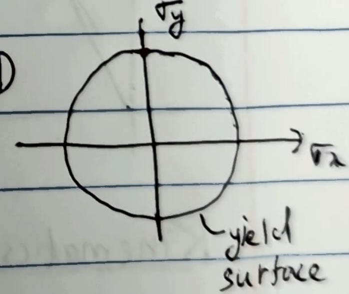
$$f(\sigma, \alpha) = |\sigma| - G(\alpha)$$

wherein

$$\boxed{f(\sigma, \alpha) \leq 0}$$

α - internal hardening variable

In 2D



Yield surface can grow (hardening) or contract (softening)

2) $\dot{\varepsilon}^p \propto \frac{\partial f}{\partial \sigma}$ (Associative flow rule)

(* Non associative in crystal plasticity)

$\dot{\varepsilon}^p = \gamma \, \mathrm{sign}(\sigma)$ (sgn - sign)

γ - consistency parameter

3) Consistency Condition

Kuhn - Tucker Conditions

$\delta_u \Pi = 0, \quad \sigma \leq \sigma_y$

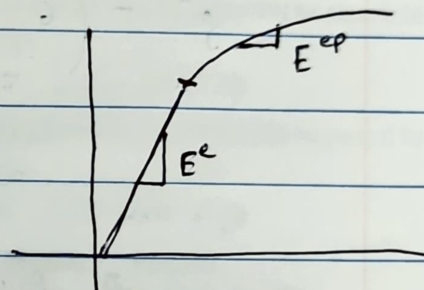
~~4.2 properties~~
~~Explicit, implicit~~

Kuhn-Tucker Conditions

$\gamma \geq 0$

$f \leq 0$ If $\gamma \geq 0$ then

$\gamma f = 0$ $\boxed{\gamma \dot{f} = 0}$ Consistency Condition



Procedure to obtain Elasto Plastic Tangent Moduli

$$\gamma \dot{f} = 0$$

If $\gamma > 0 \Rightarrow \dot{f} = 0$ $\qquad f = |\sigma| - G(\alpha)$

$$\Rightarrow \frac{\partial f}{\partial \sigma}\frac{\partial \sigma}{\partial t} + \frac{\partial f}{\partial G}\frac{\partial G}{\partial \alpha}\frac{\partial \alpha}{\partial t} = 0$$

$$\Rightarrow \operatorname{sgn}(\sigma) E(\dot{\varepsilon} - \dot{\varepsilon}^{p}) + (-1)\frac{\partial G}{\partial \alpha}\dot{\alpha} = 0$$

$$\Rightarrow \operatorname{sgn}(\sigma) E\left[\dot{\varepsilon} - \gamma \operatorname{sgn}(\sigma)\right] - \frac{\partial G}{\partial \alpha}\gamma = 0$$

$\dot{\alpha} = \gamma$ (isotropic hardening)

$$\Rightarrow E\dot{\varepsilon}\operatorname{sgn}(\sigma) - \gamma E - \frac{\partial G}{\partial \alpha}\gamma = 0$$

$$\Rightarrow \boxed{\gamma = \frac{E\dot{\varepsilon}\operatorname{sgn}(\sigma)}{E + \frac{\partial G}{\partial \alpha}}}$$

Consider

$$\dot{\sigma} = \underbrace{\frac{\partial \sigma}{\partial \varepsilon}}_{E^{ep}} \dot{\varepsilon}$$

$$\dot{\sigma} = E\left(\dot{\varepsilon} - \dot{\varepsilon}^{p}\right)$$

$$= E\left(\dot{\varepsilon} - \gamma \, \mathrm{sgn}(\sigma)\right)$$

$$= E\left[\dot{\varepsilon} - \frac{E\dot{\varepsilon}}{E + \frac{\partial G}{\partial \alpha}}\right]$$

$$\dot{\sigma} = E\dot{\varepsilon}\left[\frac{E + \frac{\partial G}{\partial \alpha} - E}{E + \frac{\partial G}{\partial \alpha}}\right]$$

$$\dot{\sigma} = \underbrace{\left(\frac{E\frac{\partial G}{\partial \alpha}}{E + \frac{\partial G}{\partial \alpha}}\right)}_{E^{ep}} \dot{\varepsilon}$$

$$\therefore \boxed{E^{ep} = \frac{E\frac{\partial G}{\partial \alpha}}{E + \frac{\partial G}{\partial \alpha}}}$$ Consistent Elasto-plastic Tangent

(2D/3D) Algorithmic Elasto-Plastic Tangent (very difficult to derive analytically)

References:

Computational Inelasticity
- Simo and Hughes